

[INDRA] Interactive Deep-dreaming Robotic Artist: Perceived artistic agency when collaborating with Embodied AI

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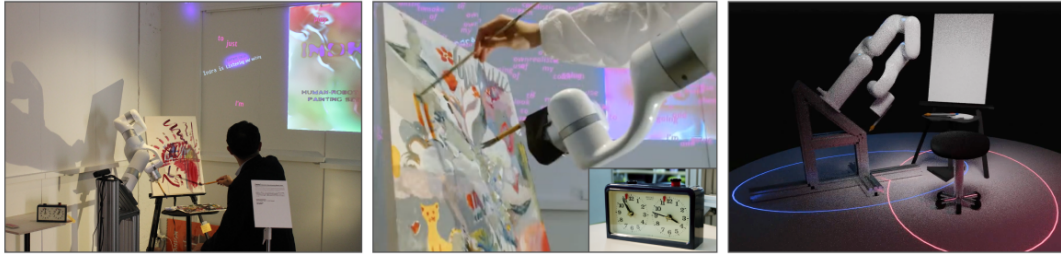


Figure 1: INDRA was exhibited over three days where participants engaged in a collaborative, turn-based painting session with a robotic arm(left), indicated with zones(right). INDRA passively listened to the painter’s ambient speech, which influenced the painting’s outcome(middle). The switch between human and robot artist is by analog button on a chess clock (middle).

Abstract

AI art generators such as DALL-E and MidJourney create artwork after man-made art styles in seconds, leveraging developments in machine learning. Traditional art, such as painting and sculpture, has long been a cornerstone of cultural expression. Human value-centered design considerations are essential when implementing AI for traditional artists, balancing automation and meaningful human control levels to produce a positive sense of creation.

This research explores artistic agency in human-computer collaborative painting through embodied AI. We present INDRA, an Interactive Deep-Dreaming Robotic Artist to enhance fine art capabilities and assist users dynamically. INDRA is designed to be an artist-friendly AI agent inspired to improve the relationship between AI and creatives.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → *Fine arts*.

Keywords

Embodied interaction, robotics, traditional painting, artistic agency

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1 Introduction

INDRA is a turn-based collaborative painting robot (physical generative AI) with human artists on a single canvas. The study aims to maintain user perceived artistic agency with an AI-integrated creative process.

The system is designed after the concept of reframing the negative perception of AI usage by the artist community by considering artist diversity and individual needs. Terms and design around AI technology imply an anthropomorphic AI being with intent and agency, negatively affecting perceptions of originality and authorship. [Epstein et al. 2023; Then et al. 2023] The real-time system considers the human artist’s intent, content, characteristics (child or adult), skill level, style, and accessibility needs into the painting by interacting in turns. Artistic Agency is measured using the Sense of Agency Scale (SoAS), and refers to the amount of meaningful human control, where the AI can independently paint but responds to the user’s input. Unlike solo artists such as Ai-Da and AICAN, INDRA’s work is produced only as a reaction to the user, fostering collaboration rather than competition.

Research on applied AI to traditional painting focuses on input photos or typed prompts [Nichol et al. 2022]. Applications to robotics such as collaborative AI art robot CoFrida [Schaldenbrand et al. 2024] are not real-time and subsequently cannot achieve quick switching between human-AI turns[Kenyon 2006].

2 System

Unlike purely digital AI art applications, INDRA tangibly interacts with a shared canvas introducing a layer of complexity and intimacy. An analog chess clock is the means of turn-switching as a

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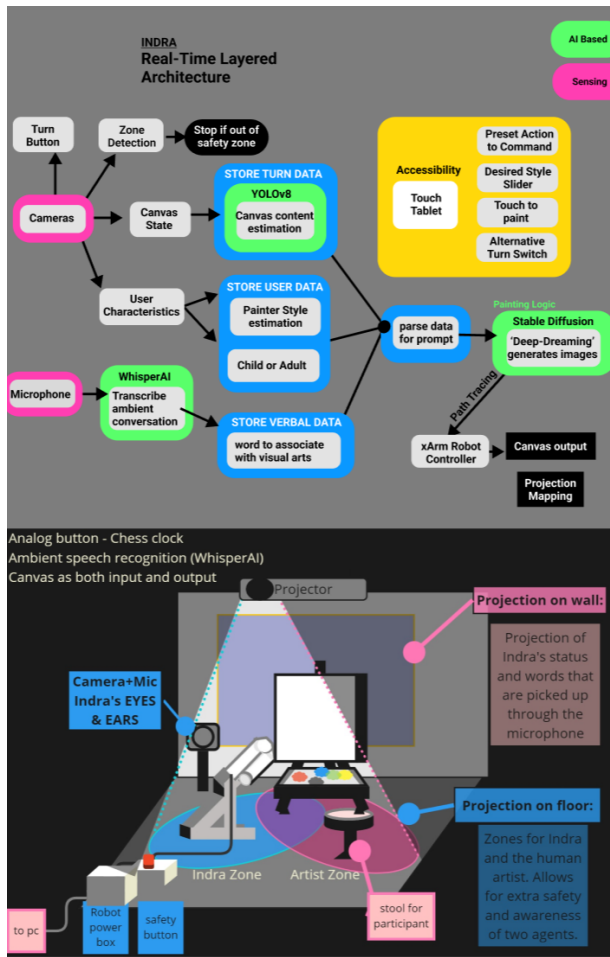


Figure 2: Diagram of INDRA Architecture

ready-made object, each turn adding to the composition (fig. 1). INDRA passively takes in data and "deep-dreams" to align with the user's artistic intentions in a traditional painting setup. INDRA accommodates to the user's needs by passively collecting information. The data is incorporated into INDRA's 'deep-dreams' of user's artistic intentions in a classic painting setup.

The system transcribes speech with OpenAI Whisper API (fig. 2). Verbal commands include dipping its brush into water. Other terms that describe the painting, intent, and feedback is inserted into a prompt dataset. The interaction occurs through the strokes on the canvas and spoken word. This limitation forces an intuitive form of communication, where the artist's intentions and the AI's responses transmit through the medium and are often ambiguous. The system accommodates users who cannot reach the physical canvas, button, or speak aloud, and optionally provides a tablet touch interface for manual control buttons, typing, and touch painting. The system is comprised of real-time components built with TouchDesigner: xArm Robot Controller, State Tracker (button, zone detection, YOLOv8), Stable Diffusion, and projection mapping.

2.1 Methodology

To explore these dynamics we use a mixed-method approach, with qualitative data gathered through participatory observation, interviews, and surveys conducted with INDRA users. Quantitative measures are evaluated with the Sense of Agency Scale (SoAS), utilized to objectively assess the artists' sense of control and authorship over the creative process [Tapal et al. 2017].

Participants are encouraged to speak their thoughts aloud during sessions, with observations documenting their interactions, inner thought processes, and reported emotions of the artists. Surveys are administered before and after the painting sessions.

3 User Experience

The robot aims to be adaptive to different participants to accommodate a range of painting styles and skill levels. Its applications can be for further accessibility use, artistic purposes, and art therapy.

- **For Traditional Artists** Paints aligned to artist's vision. Allows experienced artists to push the boundaries of their creativity and explore unexpected territories in their work.
- **For Digital Artists** An undo verbal command allows for digital artists to gain confidence in traditional medium.
- **Accessibility use** INDRA can be utilized to paint for users who are unable to grasp a brush or reach areas of a canvas by producing the user's touch drawing.
- **For children** INDRA can adapt its responses to playfully engage the narrative within the child's drawing.

4 Conclusion

This research underscores the importance of designing AI systems where human creative contribution is enhanced rather than overshadowed. INDRA serves as a promising example of how embodied AI can be integrated into traditional art forms, offering a new dimension to creative expression.

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